QP Code

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION 2026

FIFTH SEMESTER

DSC-MG5DSCBTS301- MOLECULAR BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Who demonstrated that DNA contains equal amounts of adenine and thymine and guanine and cytosine? [K/CO1]

   a) Friedrich Miescher b) Erwin Chargaff
   c) Francis Crick d) Oswald Avery

2. Which enzyme joins Okazaki fragments? [K/CO1]

   a) Primase b) DNA ligase
   c) Helicase d) Topoisomerase

3. Riboswitches commonly respond to: [K/CO4]

   a) Small molecules b) DNA polymerase
   c) Ribosomes only d) Histones

4. In prokaryotes, promoter recognition is primarily carried out by: [K/CO2]

   a) Sigma factor b) Ribosome
   c) DNA ligase d) Helicase

5. AUG generally codes for: [K/CO1]

   a) Glycine b) Methionine
   c) Valine d) Alanine

6. A major challenge in nucleic acid therapy is: [K/CO5]

   a) Delivery into target cells b) Lack of nucleotides
   c) Absence of RNA d) Lack of amino acids

7. During transcription, RNA is synthesized in the: [K/CO2]

a) 3′ → 5′ direction b) 5′ → 3′ direction

c) Both directions d) Random direction

8. Which RNA acts as an adaptor between mRNA codons and amino acids? [K/CO1]

a) mRNA b) tRNA

c) rRNA d) miRNA

9. siRNA stands for: [K/CO4]

a) Small interfering RNA b) Single intracellular RNA

c) Small initiating RNA d) Synthetic inhibitory RNA

10. A major challenge in nucleic acid therapy is: [K/CO5]

a) Delivery into target cells b) Lack of nucleotides

c) Absence of RNA d) Lack of amino acids

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. What is the basic structural unit of DNA and RNA? [K/CO1]

12. What is bacterial transformation? [K/CO1]

13. What is the function of primase? [K/CO2]

14. Differentiate between introns and exons [U/CO1]

15. Mention any two characteristics of enhancers. [K/CO2]

16. What is photoreactivation? [K/CO2]

17. Mention two contributions of Har Gobind Khorana. [K/CO1]

18. List the major stages of translation [K/CO2]

19. What is the function of aminoacyl-tRNA synthetase? [K/CO1]

20. Define a riboswitch [K/CO4]

21. What is gene therapy? [K/CO5]

22. Mention two major achievements of the Human Genome Project. [K/CO5]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Trace the major historical developments from Miescher to Watson and Crick in the discovery of DNA structure. [K/CO1]

24. Differentiate between theta, rolling circle and D-loop replication [U/CO2]

25. Explain Base Excision Repair (BER). [U/CO2]

26. Explain the process of transcription in prokaryotes. [U/CO2]

27. Describe the structure and regulation of the lac operon. [U/CO2]

28. Explain the mechanism and functions of microRNA [U/CO4]

29. Discuss the ethical, legal and social implications of the Human Genome Project. [U/CO5]

[4x5 = 20]
